DRASHTI BAMBHANIYA

Design Verification Engineer

Email: drashtiahir23@gmail.com | **Phone:** +91-9510853882 | **Github:** Drashti Bambhaniya |
LinkedIn: drashti-bambhaniya |

Education

Nirma University 8.36/10
M.Tech in Computer Science Engineering *2023 - 2025*

Gujarat Technological University 8.06/10
B.E in Information Technology *2019 - 2023*

Experience

Mirafra Technologies Aug 2025 – Present
Verification Engineer | SystemVerilog, UVM, SVA, Functional Coverage

- Built SV and UVM-based layered testbenches for an **RV32I RISC-V** processor, **AXI4-Lite slave RAM**, and **ALU design**, implementing constrained-random stimulus generation, self-checking scoreboards, and full end-to-end simulation flow across all projects.
- Structured each verification environment around UVM agent, sequencer, driver, monitor, and coverage collector architecture to ensure reusability across design blocks.
- Implemented **SystemVerilog Assertions** to validate protocol handshake behaviour and design correctness, supporting verification closure.
- Designed **functional coverage** models with targeted bins for instruction categories, protocol scenarios, corner cases, and state transitions; closed ALU testbench coverage to **98%** across 35 directed tests and root-caused **8 RTL bugs**.
- Executed simulation runs using **Mentor Questa and Synopsys VCS**, analyzed waveform outputs, and performed root-cause analysis to resolve design discrepancies across regression runs.
- Currently developing protocol-level understanding of ARM AMBA CHI - cache coherency, node architecture, transaction flows, and channel structure — through structured specification study.

NXP Semiconductors July-2024 - June-2025
Software Intern | Python, DevOps Tools, Wi-Fi, Linux Device Drivers, Debugging

- Worked extensively on various **i.MX boards** across multiple **NXP SoCs**, gaining hands-on exposure to SoC-level hardware-software integration, platform bring-up, and interface-level debugging.
- Identified and resolved **Wi-Fi performance and connectivity issues**, improving system stability and compliance with **IEEE 802.11** standards.
- Automated the **build process** for all **SoC** and **BSP** combinations using Python and Shell scripts, and integrated it into the **CI/CD pipeline** for pre-merge checks to ensure consistent driver quality.

Projects

RISC-V RV32I Processor Functional Verification | *SystemVerilog, UVM, Functional Coverage*

- Developed an **ISA** reference model covering all **RV32I** instruction types, CSR operations, branch/jump PC prediction, and load/store validation.
- Architected a layered UVM environment for a pipelined RV32I processor, wiring the reference model into the scoreboard for **instruction-level DUT** comparison.
- Designed functional coverage through constrained-random stimulus with legal opcode constraints and cross coverage across all RV32I instruction categories.

8-bit Parameterized ALU Verification | *SystemVerilog, UVM, SVA, Functional Coverage*

- Built a complete **UVM testbench** - uvm_sequence, uvm_driver, uvm_monitor, uvm_scoreboard, uvm_agent, uvm_env to verify a **25-opcode 8-bit parameterized ALU** across arithmetic, logical, and shift operations.
- Developed constrained-random stimulus with **directed sequences**, a self-checking reference-model scoreboard, and **SVA assertions** to validate correctness across all opcodes and edge cases.
- Defined functional **covergroups** with targeted bins per opcode and operand combination; **closed coverage** to **98%** across 35 directed random tests.
- Root-caused **8 RTL bugs** through **waveform** analysis in **QuestaSim and VCS**, collaborating on fixes and re-running regression to confirm closure.

Technical Skills

CPU & Architecture: RISC-V ISA (RV32I+M), Pipeline Architecture, Cache Coherency, AMBA CHI, NoC
Design & Verification: System Verilog, Verilog, UVM, Digital Design Fundamentals, Functional Coverage, Constrained-Random Verification, SVA, Waveform Debug, AXI4
Programming & Automation: C/C++, Python, Shell Scripting, Perl
EDA Tools: Synopsys VCS, Mentor Questa
Development Tools & DevOps: Git, Docker, Kubernetes, Confluence, Bitbucket, Jenkins, CI/CD Pipelines